206. Reverse Linked List

PLATFORM	DIFFICULTY	DATE
Platform: LeetCode	EASY	Solved: June 23, 2026

Problem Summary

Given the head of a singly linked list, reverse the list and return the new head.

Key Insights

- Reversal happens **in-place** — no extra list/array needed.
- You must save curr.next *before* overwriting it, or you lose the rest of the list.
- prev ends up as the new head once curr becomes null.

Section

Time Complexity : O(n) — one pass through all nodes.
Space Complexity : O(1) — only three pointers used, no extra data structures.

Approach

Iterative pointer reversal — walk through the list once, and at each node, flip its next pointer to point backward instead of forward, using a prev pointer to track the new previous node and a next pointer to save the original next node before overwriting it.

Observations

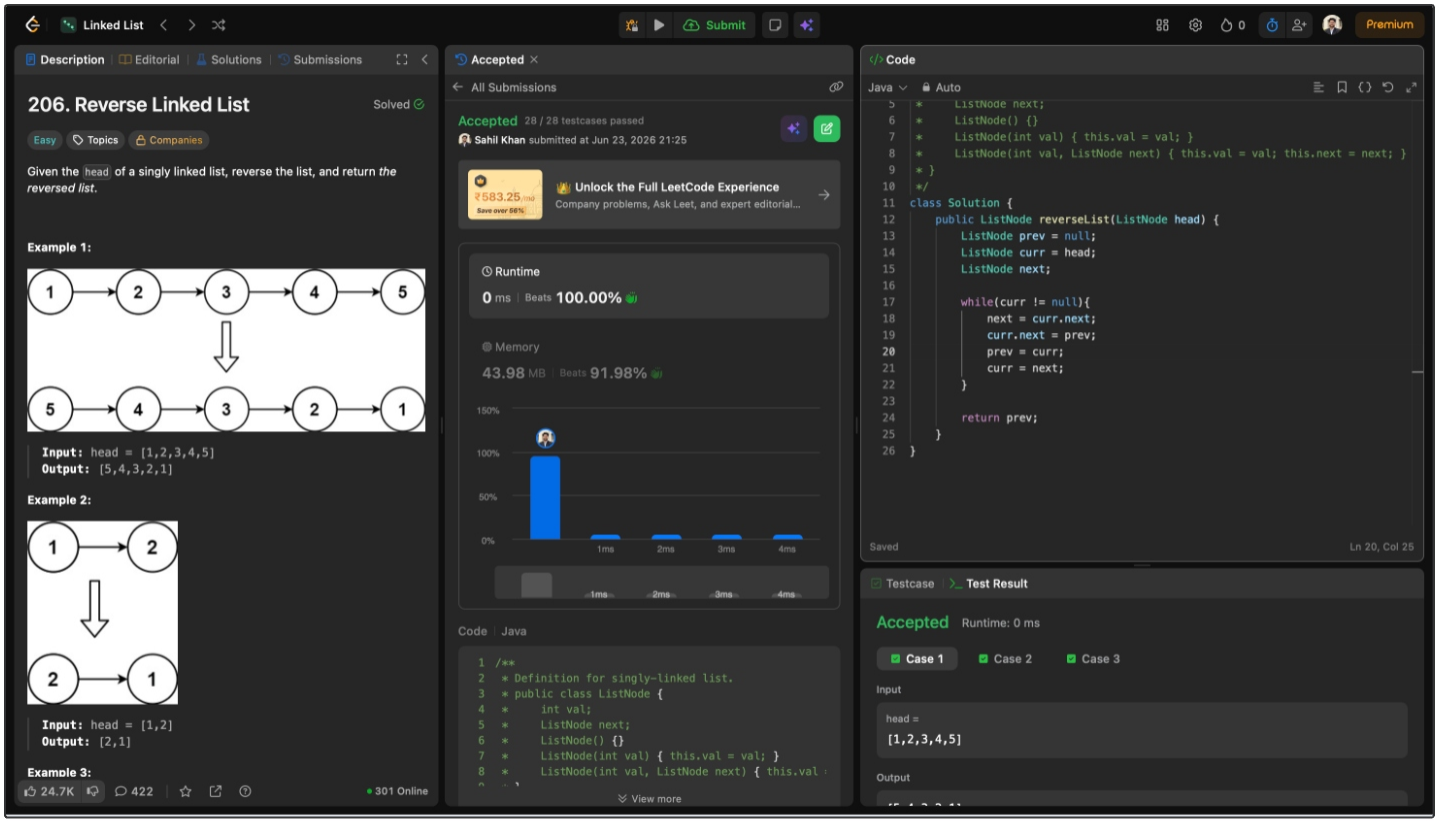
- Classic three-pointer technique (prev, curr, next) — appears in many linked list problems.
- No dummy node needed here since we're not modifying structure around a head boundary, just direction.
- Single pass, no recursion stack used (unlike the recursive variant of this problem).

Algorithm

- Initialize prev = null, curr = head.
- While curr != null:
  a. Store next = curr.next.
  b. Reverse link: curr.next = prev.
  c. Move prev = curr.
  d. Move curr = next.
- Return prev (new head).

Points to Remember

- **Three-pointer reversal pattern**: prev, curr, next — reusable for reversing sublists (e.g., reverse between positions, reverse in groups of k).
- Always cache curr.next before reassigning curr.next, or the chain breaks.
- The loop's exit condition (curr == null) means prev naturally holds the final node — that's your new head/tail depending on direction.
- This pattern generalizes to "reverse a portion of a list" by just controlling where you start/stop the loop and reconnecting boundaries afterward.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Given the head of a singly linked list, reverse the list, and return the reversed list.
Example 1:
Input: head = [1,2,3,4,5]
Output: [5,4,3,2,1]
Example 2:
Input: head = [1,2]
Output: [2,1]
Example 3:
24.7K
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Runtime
0 ms | Beats 100.00%
Memory
43.98 MB | Beats 91.98%
150%
100%
50%
0%
1ms
2ms
3ms
4ms
Code | Java
/**
* Definition for singly-linked list.
* public class ListNode {
*   int val;
*   ListNode next;
*   ListNode() {}
*   ListNode(int val) { this.val = val; }
*   ListNode(int val, ListNode next) { this.val
View more
Code
Java
Auto
*   ListNode next;
*   ListNode() {}
*   ListNode(int val) { this.val = val; }
*   ListNode(int val, ListNode next) { this.val = val; this.next = next; }
* }
*/
class Solution {
public ListNode reverseList(ListNode head) {
ListNode prev = null;
ListNode curr = head;
ListNode next;
while(curr != null){
next = curr.next;
curr.next = prev;
prev = curr;
curr = next;
}
return prev;
}
}
Saved
Ln 20, Col 25
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Input
head =
[1,2,3,4,5]
Output